

ASSIGNMENT #06

Course: CHE 341: CHEMICAL PROCESS CONTROL
Term: Spring 2026
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Due Date: Friday, April 17th TA's Mailbox

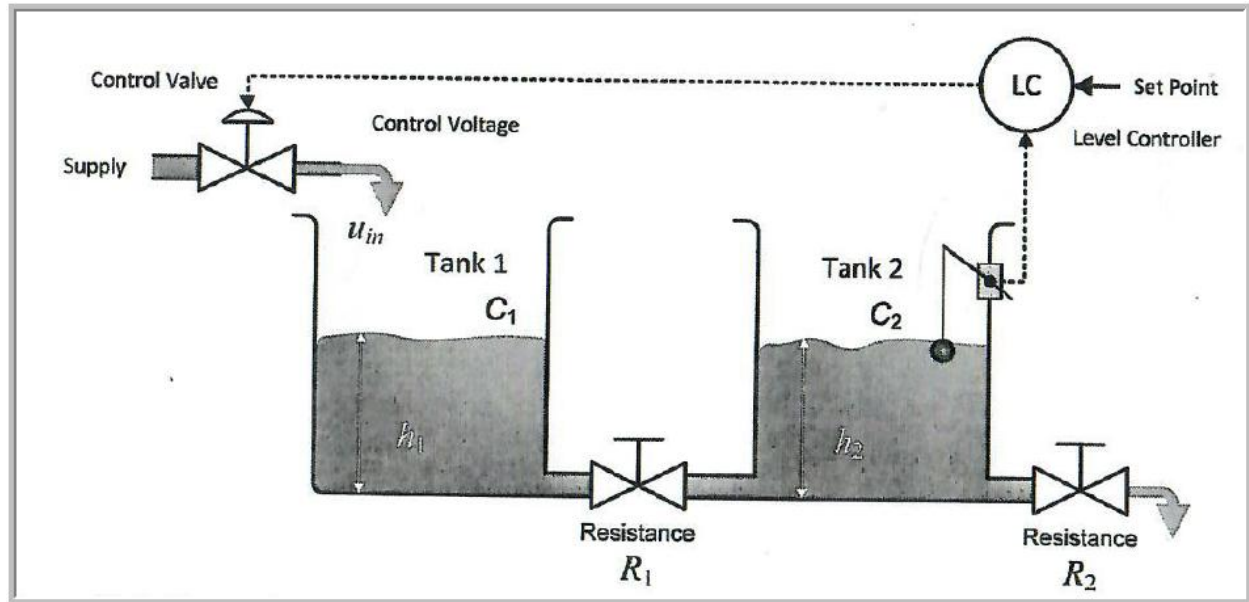
INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name in the report.
- Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time(**Closing of the Business Day**) in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem: Two-Tank Level Control System

In many chemical process applications, cascaded tank systems are used to attenuate disturbances and improve level regulation. Consider the two-tank system shown below, where liquid flows from Tank 1 into Tank 2.



A level transmitter is installed in Tank 2 and provides feedback to a PID controller, which manipulates the inlet flow rate to regulate the level in Tank 2.

The capacitance of the first and second tank is $C_1=C_2=1.0 \text{ m}^3/\text{m}$ and the outlet resistances of the first and second tank $R_1=0.5 \text{ m}/(\text{m}^3/\text{min})$ and $R_2=0.333 \text{ m}/(\text{m}^3/\text{min})$. The volumetric flow rate through the gate valves varies linearly with the pressure drop. Assume that the time delays in the control loop including the transportation lag between the first and second tank amount to around 35% of the dominant time constant of the system.

1. Derive the process and disturbance transfer functions assuming that the inlet flow acts as the manipulated variable (MV) and also as the disturbance variable (DV).
2. Design a PID controller with $\text{GM}=3$ and $\text{PM}=60$ degrees and perform a simulation for set point and disturbance change of unit size.
3. Compare your controller with:
 - a. Ziegler-Nichols Open Loop
 - b. Cohen-Coon
 - c. Ziegler-Nichols Closed Loop
 - d. Luyben-Tyreus
 - e. IMC-based tuning
 - f. Your own design

Present your results in a tabular form and compute the following metrics for both set point tracking and disturbance rejection of unit size.

1. Overshoot (if present)
2. Settling time
3. Gain Margin
4. Phase Margin
5. Maximum amount of time delay allowed before the onset of instability
6. Peak value M_s of the sensitivity function
7. Peak value M_t of the complementary sensitivity function

Include all the simulation graphs in your final report along with the Excel table shown below

	GM=3 PM=60 deg	Ziegler-Nichols Open Loop	Cohen-Coon	Ziegler-Nichols Closed Loop	Luyben-Tyreus	IMC	My Personal Controller	Selected Controller
Overshoot (if present)								
Settling time								
Gain Margin								
Phase Margin								
Maximum amount of time delay allowed before instability								
Peak Value of the sensitivity function								
Peak Value of the complimentary sensitivity function								
Controller Gain, K_c								
Controller Reset Time, τ_i								
Controller Derivative Time, τ_D								

Discuss the trade-offs observed between performance and robustness for the different controllers. Comment on which controller provides the best overall performance and justify your answer.

Assignment 6

Question 1

Let the manipulated inlet flow be $q_m(t)$, the disturbance inlet flow be $q_d(t)$, and the controlled variable be the liquid level in Tank 2. The linear valve relations are

$$q_{12}(t) = \frac{h_1(t) - h_2(t)}{R_1}$$

$$q_2(t) = \frac{h_2(t)}{R_2}$$

Applying a total volume balance to each tank gives

$$C_1 \frac{dh_1}{dt} = q_m + q_d - \frac{h_1 - h_2}{R_1}$$

$$C_2 \frac{dh_2}{dt} = \frac{h_1 - h_2}{R_1} - \frac{h_2}{R_2}$$

With the problem data,

$$C_1 = C_2 = 1.0 \text{ m}^3/\text{m}, \quad R_1 = 0.5 \text{ m}/(\text{m}^3/\text{min}), \quad R_2 = 0.333 \text{ m}/(\text{m}^3/\text{min})$$

Define the deviation variables about steady state as

$$H_1 = h_1 - h_{1,ss}, \quad H_2 = h_2 - h_{2,ss}, \quad M = q_m - q_{m,ss}, \quad D = q_d - q_{d,ss}$$

Because the model is already linear, subtracting the steady-state equations gives

$$C_1 \frac{dH_1}{dt} = M + D - \frac{H_1 - H_2}{R_1}$$

$$C_2 \frac{dH_2}{dt} = \frac{H_1 - H_2}{R_1} - \frac{H_2}{R_2}$$

Taking Laplace transforms with zero initial conditions gives

$$C_1 s H_1 = M + D - \frac{H_1 - H_2}{R_1}$$

$$C_2 s H_2 = \frac{H_1 - H_2}{R_1} - \frac{H_2}{R_2}$$

Solving for the Tank 2 level gives the common dynamic path from the inlet flow to the measured level,

$$\frac{H_2(s)}{M(s) + D(s)} = \frac{1/R_1}{C_1 C_2 s^2 + \left[C_1 \left(\frac{1}{R_1} + \frac{1}{R_2} \right) + \frac{C_2}{R_1} \right] s + \frac{1}{R_1 R_2}}$$

Therefore, the process transfer function and disturbance transfer function are

$$G_p(s) = \frac{H_2(s)}{M(s)} = \frac{1/R_1}{C_1 C_2 s^2 + \left[C_1 \left(\frac{1}{R_1} + \frac{1}{R_2} \right) + \frac{C_2}{R_1} \right] s + \frac{1}{R_1 R_2}}$$

$$G_d(s) = \frac{H_2(s)}{D(s)} = \frac{1/R_1}{C_1 C_2 s^2 + \left[C_1 \left(\frac{1}{R_1} + \frac{1}{R_2} \right) + \frac{C_2}{R_1} \right] s + \frac{1}{R_1 R_2}}$$

Substituting the numerical values of C_1 , C_2 , R_1 , and R_2 gives

$$G_p(s) = G_d(s) = \frac{2}{s^2 + 7s + 6} = \frac{2}{(s + 1)(s + 6)}$$

The slower pole is at $s = -1$, so the dominant time constant is 1 min. The assignment states that the total loop delay is 35% of the dominant time constant, so

$$\theta = 0.35 \text{ min}$$

Including the delay in the measured-variable path gives the transfer functions used for controller design,

$$G_p(s) = \frac{2e^{-0.35s}}{(s + 1)(s + 6)}$$

$$G_d(s) = \frac{2e^{-0.35s}}{(s + 1)(s + 6)}$$

The final result is that the two-tank plant behaves as an overdamped second-order process with time delay, and the manipulated-variable and disturbance-variable transfer functions are identical because both enter through the same inlet-flow path.

Question 2

The controller is written in the standard non-interactive PID form,

$$G_c(s) = K_c \left(1 + \frac{1}{\tau_I s} + \tau_D s \right)$$

For CHE Calculator, it is convenient to rewrite the plant in standard self-regulating SOPTD form as

$$G_p(s) = \frac{(1/3)e^{-0.35s}}{(1.0s + 1)(0.1667s + 1)}$$

Since the controlled variable is the level in Tank 2 and the manipulated variable is the inlet flow, the loop is taken as reverse acting with the usual error definition

$$E = SP - PV$$

The controller is tuned in the Bode worksheet to satisfy the robustness targets

$$GM = 3, \quad PM = 60^\circ$$

Using the open-loop transfer function $L(s) = G_c(s)G_p(s)$, the calculator is adjusted until the crossover conditions are satisfied,

$$|L(j\omega_g)| = 1, \quad \angle L(j\omega_g) = -120^\circ$$

$$|L(j\omega_c)| = \frac{1}{3}, \quad \angle L(j\omega_c) = -180^\circ$$

The resulting controller settings are

$$K_c = 5.253, \quad \tau_I = 1.152 \text{ min}, \quad \tau_D = 0.149 \text{ min}$$

so the final controller becomes

$$G_c(s) = 5.253 \left(1 + \frac{1}{1.152s} + 0.149s \right)$$

For the required closed-loop simulations, the set-point and disturbance transfer functions are

$$\frac{PV(s)}{SP(s)} = \frac{L(s)}{1 + L(s)}$$

$$\frac{PV(s)}{D(s)} = \frac{G_d(s)}{1 + L(s)}$$

The unit set-point test is run with a step change in SP while keeping the disturbance at zero. The unit disturbance test is run with the set point held constant while applying a unit disturbance through the inlet-flow path. Since the inlet flow acts as both the manipulated variable and the disturbance variable, the same dynamic model is used for both paths.

The practical summary is to use the Bode worksheet to satisfy $GM = 3$ and $PM = 60^\circ$, then transfer the final PID settings into the simulation worksheet and run the unit set-point and unit disturbance cases.

Question 3

Comparison basis. All controllers were compared on the same full plant model so that the differences in the results reflected tuning only and not a change in the process representation. The exact plant retained for the final comparison was the delayed second-order model used earlier in the report.

$$G_p(s) = \frac{2e^{-0.35s}}{(s+1)(s+6)} = \frac{0.333 e^{-0.35s}}{(s+1)(0.1667s+1)}$$

Tuning basis. The GM = 3, PM = 60° controller was kept from Part 2 and therefore required no model reduction. Ziegler–Nichols open loop, Cohen–Coon, and IMC were computed from a first-order-plus-time-delay approximation, while Ziegler–Nichols closed loop and Tyreus–Luyben were based on the ultimate gain and ultimate period. The personal controller was selected directly.

FOPTD approximation. For the methods written in FOPTD form, Skogestad's half-rule was used to reduce the exact SOPTD model. With $\tau_1 = 1.0$ min, $\tau_2 = 0.1667$ min, and $\theta = 0.35$ min, the reduced constants were:

$$\tau = \tau_1 + \frac{\tau_2}{2} = 1.0833$$

$$\theta * = \theta + \frac{\tau_2}{2} = 0.4333$$

$$G_p(s) \approx \frac{0.333 e^{-0.4333s}}{1.0833s + 1}$$

Tuning relations. The controller constants were then obtained from the standard formulas associated with each method.

Ziegler–Nichols open loop

$$K_c = \frac{1.2}{\alpha K_p} \frac{\tau}{\theta}, \quad \tau_I = 2\theta, \quad \tau_D = \frac{\theta}{2}$$

Cohen–Coon

$$K_c = \frac{1.35}{\alpha K_p} \left(\frac{\tau}{\theta} + 0.185 \right), \quad \tau_I = 2.5\theta \left(\frac{\tau + 0.185\theta}{\tau + 0.611\theta} \right), \quad \tau_D = 0.37 \left(\frac{\tau\theta}{\tau + 0.185\theta} \right)$$

IMC

$$K_c = \frac{\tau + \theta/2}{K_p(\tau_c + \theta/2)}, \quad \tau_I = \tau + \theta/2, \quad \tau_D = \frac{\tau\theta/2}{\tau + \theta/2}$$

Ziegler–Nichols closed loop

$$K_c = 0.60K_u, \quad \tau_I = 0.50P_u, \quad \tau_D = 0.125P_u$$

Tyreus–Luyben

$$K_c = 0.45K_u, \quad \tau_I = 2.2P_u, \quad \tau_D = 0.1587P_u$$

Evaluation procedure. After the controller constants were calculated, every controller was tested on the same full plant using a unit set-point change and a unit disturbance change. Overshoot and settling time were taken from the time responses. The frequency-domain comparison used the open-loop transfer function and the standard robustness relations.

$$L(j\omega) = Gc(j\omega)Gp(j\omega)$$

$$\Delta\theta_{max} = \frac{PM \pi}{180\omega_g}$$

$$M_S = \max_{\omega} \left| \frac{1}{1 + L(j\omega)} \right|$$

$$M_T = \max_{\omega} \left| \frac{L(j\omega)}{1 + L(j\omega)} \right|$$

$$Dead - Time Ratio = \frac{\Delta\theta_{max}}{\theta}$$

Concise comparison

The GM = 3, PM = 60° controller provided the reference case. Its response was well balanced, with modest overshoot, reasonable settling time, and robustness values that were acceptable by design.

Ziegler–Nichols open loop and Cohen–Coon were the most aggressive controllers. Both produced fast responses, but the oscillation and overshoot were much larger than the other methods, and their gain and phase margins were comparatively weak. Cohen–Coon was the least desirable of the two because it combined the largest oscillation with the poorest robustness.

Ziegler–Nichols closed loop improved on the open-loop empirical methods. The response was still somewhat aggressive, but it was better damped and more acceptable than Ziegler–Nichols open loop and Cohen–Coon.

Tyreus–Luyben was the most conservative controller. It gave the largest stability margins and the greatest delay tolerance, but that robustness came at the cost of much slower set-point and disturbance responses.

IMC gave the best overall compromise. Its responses were smooth and well damped, the overshoot remained small, and the robustness measures stayed strong. Even though its constants were obtained from an FOPTD approximation, it performed very well when tested on the full plant.

The personal controller also performed well and was the closest alternative to IMC. Its behavior supported the conclusion that derivative action was not especially important for this delayed level-control process and that a well-tuned PI controller could still provide strong overall performance.

For disturbance rejection, settling time was the more useful indicator than overshoot alone because a controller could show a small positive peak while still having a noticeable negative excursion before returning to zero. Below is a summary of values for each method.

	GM=3 PM=60 deg	Ziegler-Nichols Open Loop	Cohen-Coon	Ziegler-Nichols Closed Loop	Luyben-Tyres	IMC	My Personal Controller	Selected Controller
present), %	5.48	38.33	56.45	49.78	-	1.58	4.05	4.05
min	0.725	1.653	2.454	1.865	1.781	0.407	0.478	0.478278522
	3	1.628	1.47	1.572	2.098	2.558	3.142	3.142
degree	60.02	43.2	28.845	34.985	79.812	64.122	61.352	61.352
ount of time delay allowed before instability, min	1.048	0.62	0.515	0.561	1.059	1.018	1.1	1.1
the sensitivity function	1.624	2.773	3.575	3.041	1.978	1.72	1.59	1.59
the complimentary sensitivity function	1.009	1.891	2.765	2.206	1.048	1	1	1
Controller Gain, Kc	5.253	9.501	10.678	10.053	6.525	6.025	5	5
Controller Reset Time, τ_i , min	1.152	1.228	1.184	1.068	4.032	1.284	1.167	1.167
Controller Derivative Time, τ_D , min	0.149	0.197	0.142	0.171	0.253	0.199	0.143	0.143

From this summary, I would choose My Personal Controller because it stays very close to the required robustness target, gives the largest delay tolerance, and has the lowest M_s among the robust choices, while still keeping set-point overshoot and settling time reasonable.

Conclusion

The comparison shows a clear pattern. The most aggressive empirical methods were fast but weak in robustness, the most conservative method was robust but slow, and the best overall controllers were IMC, the personal controller, and the GM = 3, PM = 60° reference design. Among these, IMC provided the strongest overall compromise and was therefore the most appropriate final recommendation for the two-tank level-control process.

References

CHE Calculator V13.1

CHE 341 Textbook